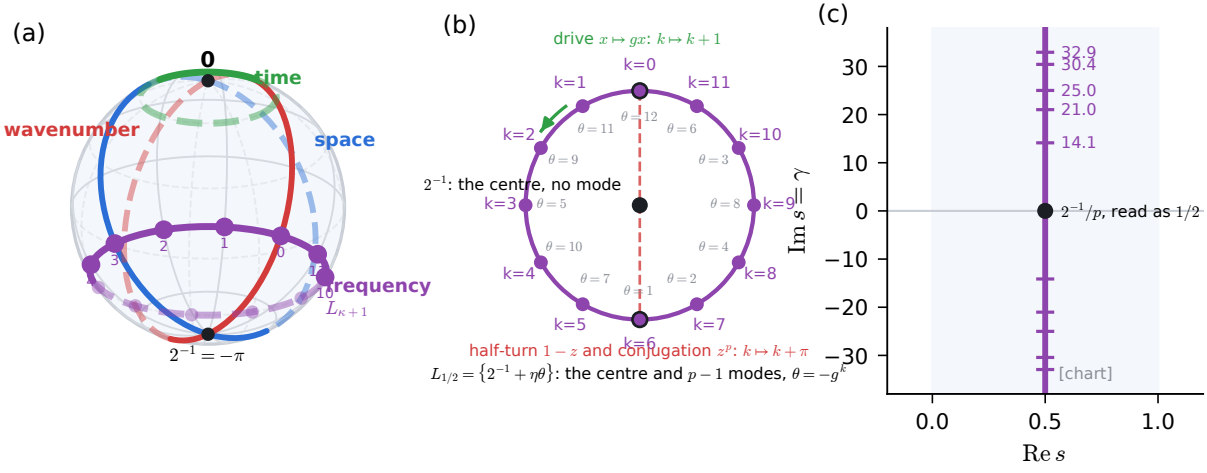

(a)
0
time
wavenumber
space
frequency
$L_{\kappa+1}$
$2^{-1} = -\pi$
(b)
drive $x \mapsto gx$: $k \mapsto k+1$
k=0
k=1
k=11
k=2
k=10
k=3
k=9
k=4
k=8
k=5
k=7
k=6
$\theta=12$
$\theta=11$
$\theta=6$
$\theta=9$
$\theta=3$
$\theta=5$
$\theta=8$
$\theta=10$
$\theta=4$
$\theta=7$
$\theta=2$
$\theta=1$
2^{-1}: the centre, no mode
half-turn $1-z$ and conjugation z^p: $k \mapsto k+\pi$
$L_{1/2} = \{2^{-1} + \eta\theta\}$: the centre and $p-1$ modes, $\theta = -g^k$
(c)
32.9
30.4
25.0
21.0
14.1
$2^{-1}/p$, read as $1/2$
[chart]
$\mathrm{Im}\, s = \gamma$
$\mathrm{Re}\, s$
30
20
10
0
−10
−20
−30
0.0
0.5
1.0